

Introduction to Computer Simulation

Home Work 4

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November 17, 2025

Question 1

1.a)

The normalization constant of choice is the maximum of X_i , so that no matter the range of the non normalized function, it will always be in $[0, 1)$.

1.b)

Plotting the time average of X_i we get:

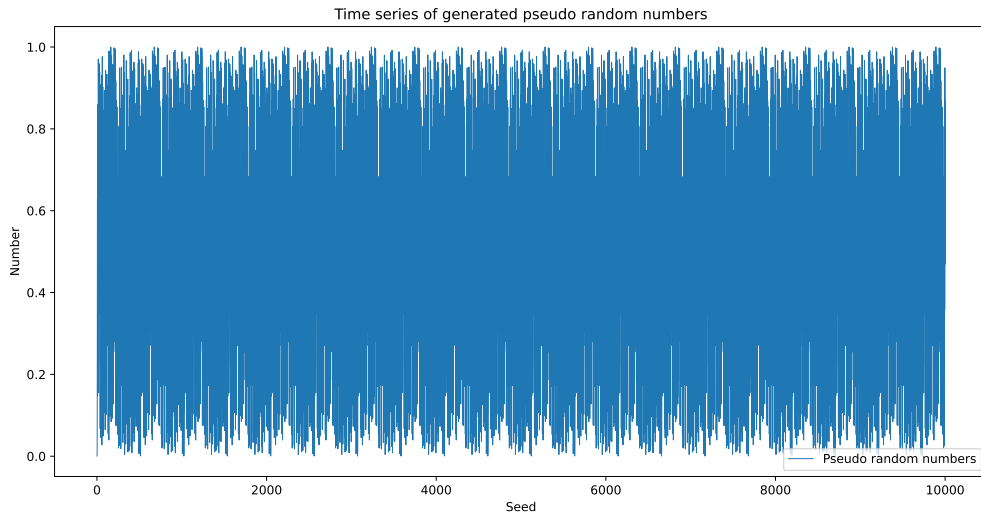


Figure 1: Plot of the pseudo-random numbers

The period given 10^4 iterations is, $\tau = 512$, in practice this means that only the first 512 numbers will seem random, any more than that and the repetition becomes clear.

1.c)

Computing $\bar{X}$ & $\hat{\sigma}_X^2$ for different N values we get the following table:

	$\bar{X}$	$\hat{\sigma}_X^2$
$N = \tau$	0.50	0.08
$N = 10\tau$	0.50	0.08
$N = 100\tau$	0.10	0.06

Table 1: values of $\bar{X}$ & $\hat{\sigma}_X^2$, for different N

We can see that not only the values for $N = \tau$ and $N = 10\tau$ stay the same, but the values for higher N , get worse, this is due the periodic nature of the random numbers, also note that for 10^4 iterations we

only have 20 distinct periods, that is why the values for $N = 100\tau$ presents a different behavior to the others.

1.d)

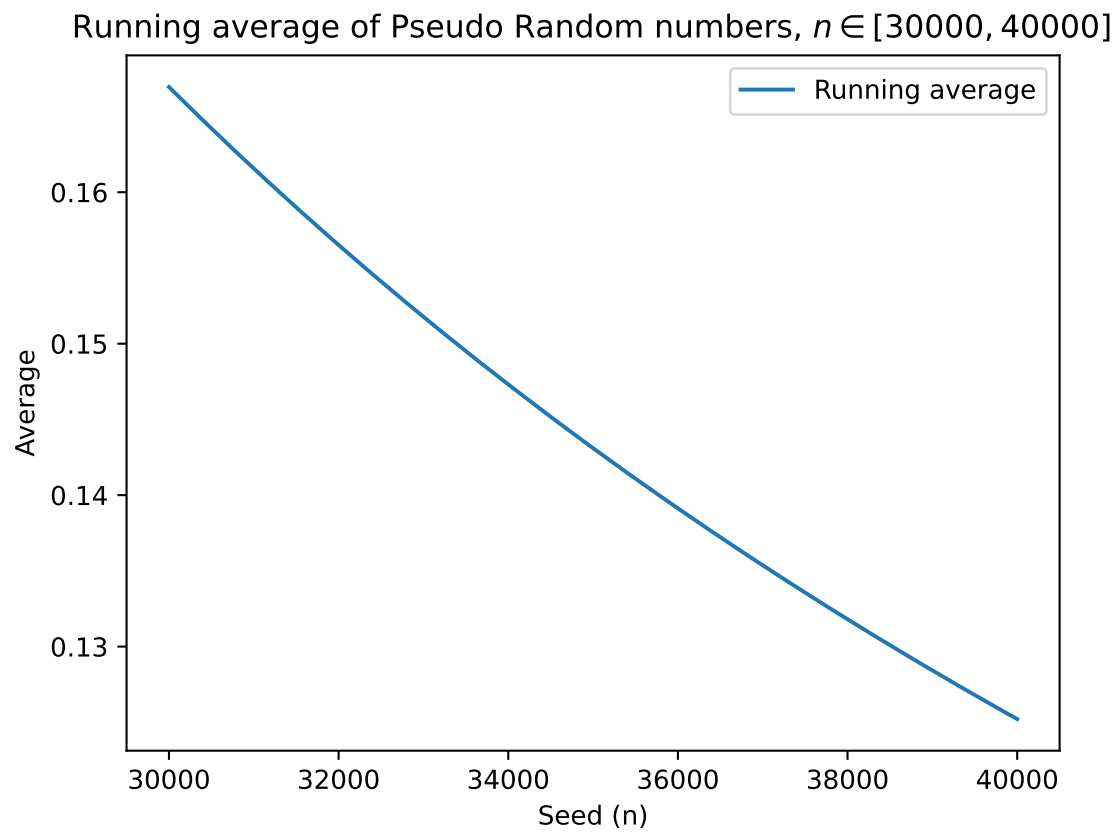


Figure 2: Plot of the running average

1.e)

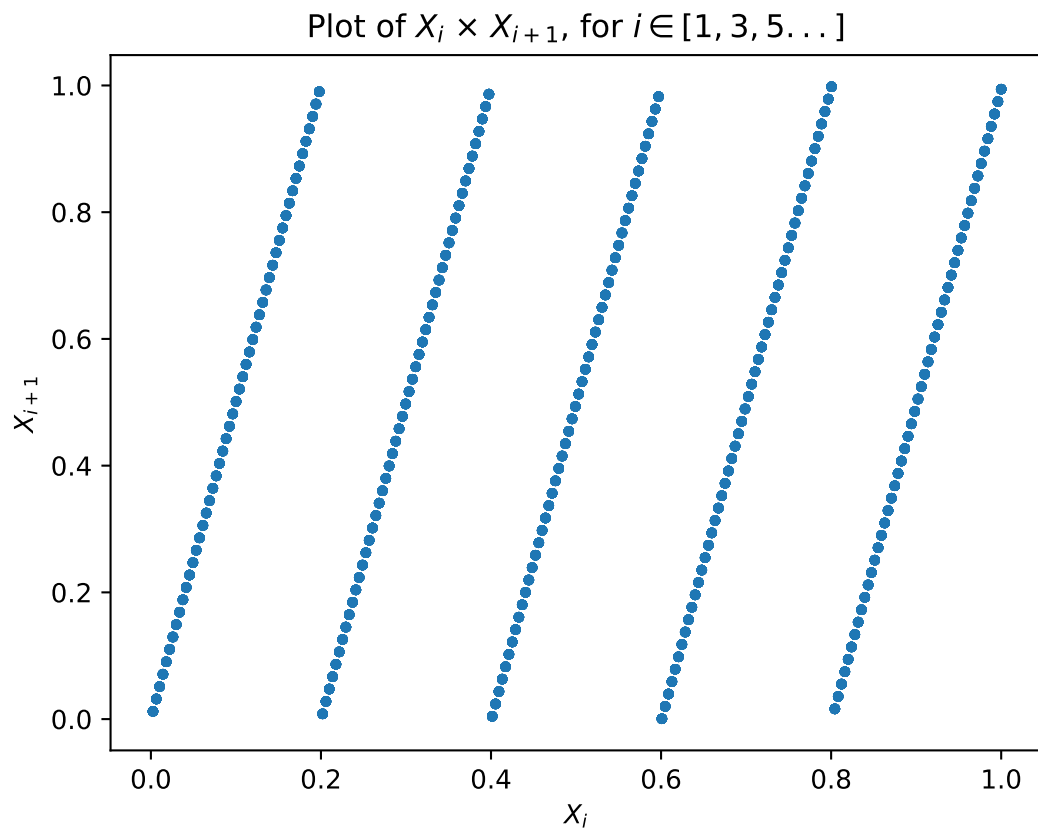


Figure 3: Plot of the running average

Question 2

Graphing the time evolution of our stochastic Ising model simulation for different initial parameters:

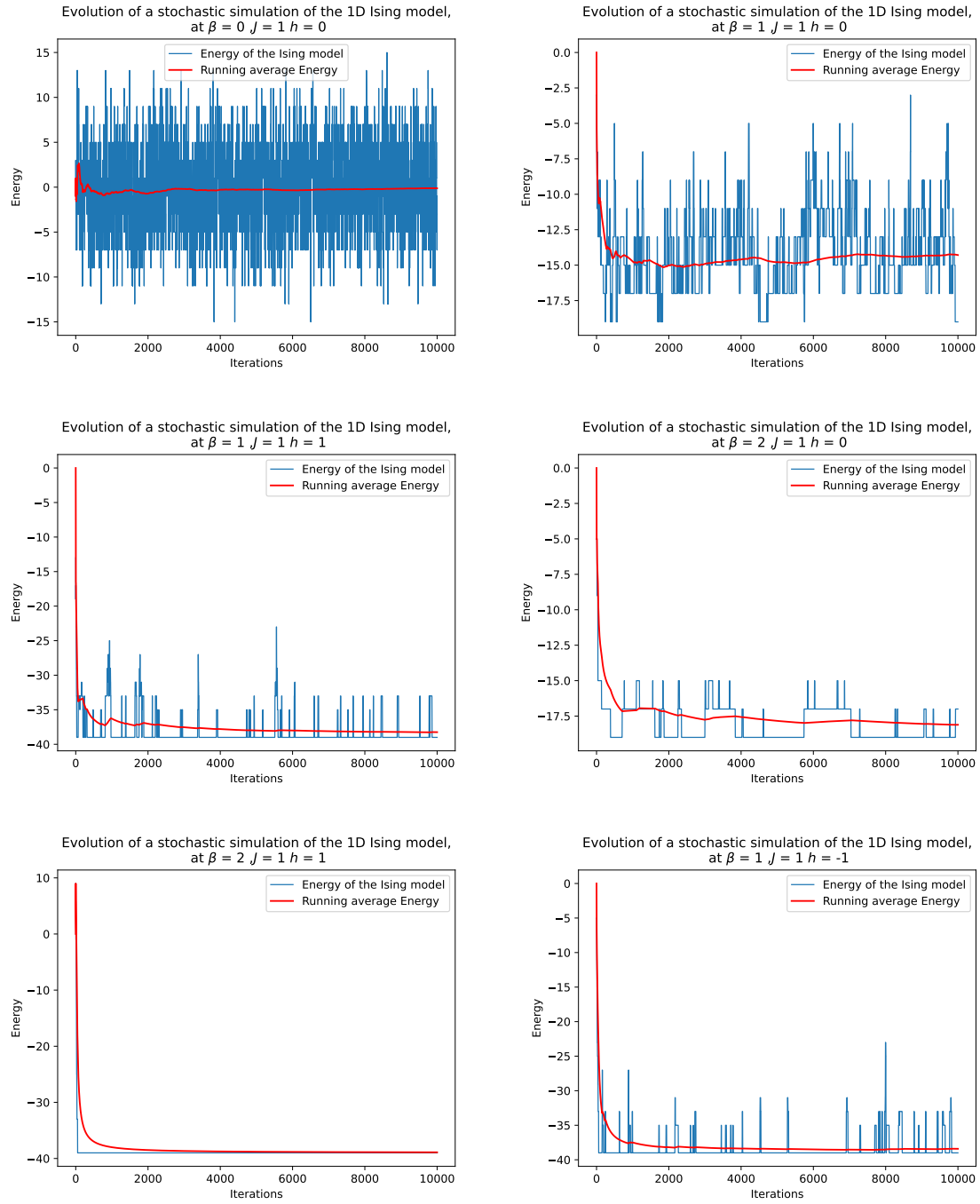


Figure 4: Time evolution for different initial conditions

We can also make a table to see what value the equilibrium energy is going to approach:

β	J	h	E
0	0	0	0.075
1	0	0	-14.288
1	1	1	-38.320
2	1	1	-38.900
2	1	0	-18.112

Table 2: Average energies for different parameter sets.

Specific heat and magnetization as a function of β :

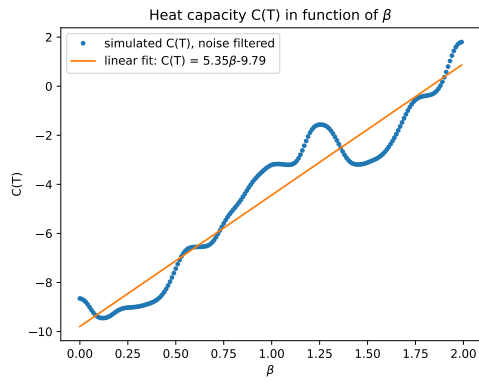


Figure 5: Specific Heat

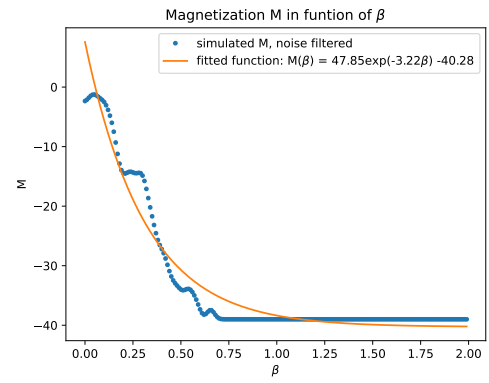


Figure 6: Magnetization